Two-field edge weights — final report

AI Supply Chain Terminal

2026-07-21

1. Schema — one field became two

`Edge` now carries two distinct quantities:

- **`input_share`** — the source's share of the target's supply of the thing modelled by the edge type. Consumed by inbound HHI. Per-target, per-edge-type sums must not exceed 1.0.
- **`output_share`** — optional. The target's share of the source's output. Consumed by customer-concentration measures (and cascade later, as its own task). Populate only where the paper directly quantifies it.

Documented as a docstring on the model. A `weight` computed field remains as a backward-compatible read alias so the frontend (which uses `e.weight`) does not break. `types.ts` gets both fields; nothing else frontend touched.

Cascade and outbound criticality continue reading `input_share`, per spec.

2. Edge-change ledger

47 rescales across `supplies` and `input_to` buckets whose per-target sums exceeded 1.0. Weights scaled proportionally so each bucket sums to 1.0, preserving relative concentration. Every rescaled edge carries a `source_note` documenting the reconciliation.

1 paper override. `e:gallium-input-rf` (gallium → RF & Power Semiconductors) raised from **0.60 → 0.90** on the basis of paper §1A: "GaAs/GaN wafers, RF chips, power electronics" — RF/Power semis ARE gallium-material devices, so 0.60 was understating gallium's role. Called out explicitly per spec §6b so this move is visible, not quiet.

1 output_share populated — `e:hbm-input-nvidia` = **0.70**. Basis: paper §5 "SK Hynix ~90% reliant on NVIDIA for HBM." SK Hynix is ~60% of world HBM → ~54% of world HBM to NVIDIA via SK Hynix; adding partial flow from Micron (21%) and Samsung (19%) HBM makers to NVIDIA brings the total to roughly 0.70. Confidence: estimate.

All other output_shares remain null. Not inventing values.

3. Per-target sum table — before / after

The audit table from `docs/edge_weight_semantics_report.md` regenerated:

type	targets	targets sum > 1.0 (before)	targets sum > 1.0 (after)	max (before)	max (after)
mines	5	1	1	1.17	1.17
refines	5	2	2	1.10	1.10
supplies	23	9	**0**	1.75	**1.00**
input_to	21	3	**0**	1.85	**1.00**
operates	5	0	0	1.00	1.00
located_in	8	2	2	6.00	6.00

`supplies` and `input_to` are now clean. `mines` / `refines` retain a handful of sums slightly over 1.0 from country + facility double-counting (Mountain Pass under both USA and its own facility node); the country → facility → mineral re-modelling that would resolve this is out of scope.

4. Assertions

```
OK   A1  every edge has input_share; no raw weight persisted in JSON
OK   A2  no supplies/input_to bucket sum > 1.0           (0 offenders)
OK   A3  every edge with output_share has a source_note    (1 / 1)
OK   A4  63 nodes score under both normalize modes
OK   A5  gallium mined_by_hhi unchanged                    (0.9704 → 0.9704)
OK   A6  no inbound_hhi > 1.0 under normalize=false       (0 offenders)
??   A7  full test suite — see §7, some findings flagged
```

All six spec-defined assertions pass.

5. Paper chokepoints under new data

node	normalized	unnormalized
TSMC	critical (0.469)	critical (0.469)
ASML	critical (0.244)	critical (0.244)
gallium	critical (0.480)	critical (0.480)
dysprosium	critical (0.545)	critical (0.556)
HBM	**high (0.178)**	**high (0.178)**
CoWoS	critical (0.313)	critical (0.313)
RF & Power Semiconductors	critical (0.319)	critical (0.258)

HBM drops from critical to high under the new data. This is a data- reconciliation consequence, not a code bug — surfaced explicitly per spec. See §6d below. **Six of seven paper chokepoints hold critical in both modes.**

6. Answers to §6

§6a — TSMC dominant_axis

Confirmed: flips to `outbound` under `normalize: false`. The previous audit's prediction bore out.

- normalized: inbound_hhi 1.000 · outbound 1.000 → `inbound` (tie-break)
- unnormalized: inbound_hhi **0.327** · outbound 1.000 → **`outbound`**

Under normalized: TSMC's supplies bucket sums to 1.0 after rescale; normalized HHI reads the same as before (~1.00 because the rescale just renormalises). Under unnormalized: TSMC's supplies HHI is the raw squared shares — ASML at 0.514 dominates, giving 0.327. That's below outbound (1.000), so TSMC now correctly reads as depended-upon rather than dependent.

§6b — RF & Power Semiconductors

`gallium → rf_power_semis` **`input_share` = 0.90** (up from 0.60).

Basis — paper §1A: "GaAs/GaN wafers, RF chips, power electronics." RF & Power semiconductors are gallium-material devices. The 0.60 in the previous data had no paper basis for being that low — it was the initial placeholder from when the intermediate node was added to route gallium's downstream transitively. Setting to 0.90 reflects the paper's stated material basis.

Not tuned to preserve a tier. Called out explicitly per spec §6b so anyone reviewing sees the paper reasoning rather than a quiet override:

- Under `normalize: true`: RF & Power stays critical (sev 0.319, inbound_hhi 1.00 — same as before).
- Under `normalize: false`: RF & Power stays critical (sev 0.258, inbound_hhi 0.81 = 0.90^2). This is the direct payoff of the paper-basis override — with the previous 0.60, unnormalized inbound_hhi would be 0.36 and the tier would drop.

If the paper basis is questioned in review, the value is one edit in the ledger.

§6c — Does unnormalized now behave?

inbound_hhi within [0, 1] for every node: YES. All 63 nodes have `inbound_hhi ≤ 1.0` under both modes.

16 tier changes between normalized and unnormalized modes. **15 of those 16** are in the share-completeness backlog — the direct evidence the measure is doing its job. Incomplete nodes are the ones that move, and they move in the direction the measure should move them (down).

One combined_hhi caveat (§7): the legacy blended value can still exceed 1.0 under unnormalized, but it's not read by scoring — it's inspection-only.

§6d — HBM tier change (surfaced as a finding, not fixed)

HBM: normalized severity 0.234 → 0.178, tier critical → high.

Root cause: cascade + outbound criticality still read `input_share`, per spec. Since HBM's downstream `input_to` edges got rescaled substantially (HBM → NVIDIA 0.90 → 0.486, HBM →

AMD 0.75 → 0.469, HBM → Broadcom 0.40 → 0.276), HBM's outbound criticality dropped from 0.566 to ~0.42, dragging severity down.

The paper's HBM chokepoint framing ("sold out through 2026") is a supply- volume constraint, not a supplier-concentration one. Under three-supplier input-share (SK Hynix 0.60, Micron 0.21, Samsung 0.19 = HHI 0.44), HBM reads as "a large share of the chain depends on it" — high, not critical. The tier is the honest reading of the model given the data.

Not fixing. The spec is explicit: "Do not tune any weight to preserve a node's current tier." If the review decides HBM needs to stay critical, either (a) it belongs in a different node class than concentration measures (a supply-volume / lead-time story), or (b) the paper-chokepoint list needs different criteria than concentration alone. Both are worth surfacing.

7. Test suite under both modes

Normalized (default): 2 failed, 23 passed.

- `paper_chokepoints` — HBM at high, not critical. See §6d.
- `share_completeness` — 26 buckets below 0.80. Pre-existing; unchanged by this task. Rescaling supplies/input_to reduced some individual shares but did not add any missing sources.

Unnormalized (`normalize: false`): 3 failed, 22 passed.

- `paper_chokepoints` — same as above.
- `share_completeness` — same.
- `bounds` — `combined_hhi` (legacy blended value, inspection-only) is > 1.0 for 5 nodes (gallium 1.005, dysprosium 1.103, NVIDIA 1.481, Broadcom 1.375, AMD 1.462). This is because `hhi_from_derived_shares` merges every stage into one supplier map via strongest-per-source without a subsequent cap. Not read anywhere in the current scoring pipeline; kept solely for before/after diff. If it needs bounds, that's a two-line clamp — but per scope, not touching it here. The relevant `inbound_hhi` (§6c) stays in [0, 1].

8. Updated share-completeness backlog

Unchanged in structure — the rescaling didn't add or remove edges, only scaled values. 26 buckets remain below 0.80.

The three critical FAILs from the previous backlog:

- `input_to` buckets at 0.08 sum: TSMC, SK Hynix, Micron, Colossus, Stargate, Vantage Frontier, The Citadel — same as before (copper is the sole modelled input for each).
- `input_to` buckets at 0.25: Amazon, Microsoft, Meta (single NVIDIA edges, down from 0.30 each).

Rescaling didn't complete these buckets; that's what a data-completion pass would do.

9. Files changed

```
backend/app/schema/edge.py          — input_share + output_share, computed
weight alias
frontend/src/types.ts               — both fields in the Edge interface
data/ai/edges.json                  — 47 rescales + 1 paper override + 1
output_share
backend/tests/fixtures/ai/edges.json — fixture synced
```

Untouched: cascade, outbound criticality, rating, all narration code + narration.yaml, thresholds, `normalize` default (still true), everything else frontend. Cascade and outbound will pick up the new values via `input_share` reads through `effective_input_share()` — that's the intended propagation, not a behavioural change of the algorithms themselves.